

8. Specifications and Operational Explanation

8.1 Specifications

Product name	: PM2.5 monitor
Model name	: FPM-388
Element to be measured	: Fine particulate matter (PM2.5) in the air
Measurement method	: Beta (β) ray absorption method.
Measurement range	: 0 to 1000 $\mu\text{g}/\text{m}^3$
Display unit	: $\mu\text{g}/\text{m}^3$ (mg/m ³ at calibration of calibration film)
Material sampling	: Collection method Filtration method Sizing method Cyclone method (VSC cyclone) Suction flow rate 16.7L/min (actual flow rate) Used filter paper Filters made of polytetrafluoroethylene (rolled, replaced monthly) Material of contacting sample air ... Stainless steel, anodized aluminum Collection time 59 minutes
Performance	: Linearity (indication error) $\pm 3\%$ or less (depending on calibration film) Repeatability (zero) $\pm 2\%$ FS or less (depending on calibration film) Repeatability (span) $\pm 2\%$ or less (for calibration film value) Zero drift $\pm 2\%$ FS or less/24h (depending on calibration film) Span drift $\pm 3\%$ or less/24h (depending on calibration film) Temporal stability of flow rate $\pm 4\%$ /24 hours
Minimum display value	: 0.1 $\mu\text{g}/\text{m}^3$ when the display unit is set to $\mu\text{g}/\text{m}^3$
Recording capacity of CF card	: For 3 years (1-hour value, calibration log data, alarm log data, etc.) * Use the genuine product of our company (256MB).
Allowable ambient temperature and humidity (performance guarantee)	: 0 to 40°C 85%RH or less (no condensation)
Warm-up time	: 3 hours
Power supply	: 220VAC, 50/60Hz (same as power supply for pump)

Power consumption	: Max. approx. 400VA, average 250W (including power supply for pump)
Mass	: Main unit 20kg Pump unit 12kg PM10 inlet + PM2.5 sizer 3.2kg (excluding down tube)